Global Technology Companies: K-Means Clustering Analysis

Fundamentals, valuation, risk and momentum segmentation | 2026 dataset

Python | Pandas | scikit-learn | Seaborn | K-Means

1. Executive Summary

This project groups global technology companies into financially similar peer segments using K-Means clustering. Seven features capture company size, innovation intensity, profitability, valuation, market risk and price momentum. After winsorising extreme observations and standardising all features, the number of clusters was selected by maximising the silhouette score over k = 2 to 15.

Model result	Value	Interpretation
Selected number of clusters	5	Highest silhouette score among the tested values
Best silhouette score	0.353	Moderate separation; useful but overlapping segments
Clustering method	K-Means	Distance-based segmentation on standardised features
Outlier treatment	1st–99th percentile clipping	Reduces the influence of extreme financial ratios
Reproducibility	random_state = 41	Produces stable assignments across reruns

Main conclusion. The model identifies five distinct financial archetypes: below-average/low-momentum companies, mega-cap profitable leaders, high-momentum higher-risk companies, R&D-intensive low-margin companies, and expensive high-risk innovation companies. The clusters are economically interpretable, but the silhouette score of 0.353 indicates only moderate separation, so the output should support peer screening rather than replace company-specific research.

2. Project Objective and Data

The objective is to segment technology companies using multiple financial dimensions simultaneously. Unlike a one-factor screen, clustering can reveal companies with similar combinations of size, profitability, innovation, valuation, risk and price performance. The source dataset is the “Global Tech Giants Fundamentals and Valuations 2026” dataset supplied through Kaggle.

Dataset source: https://www.kaggle.com/datasets/kanchana1990/global-tech-giants-fundamentals-and-valuations-2026

Feature	Financial meaning	Why it matters
Market_Cap_B	Company size in billions	Separates mega-cap leaders from smaller firms
R&D_to_Revenue_Pct	Innovation spending intensity	Captures reinvestment in technology and future products
Operating_Margin_Pct	Operating profitability	Measures business-model economics and cost discipline
Trailing_PE	Current earnings valuation	Shows the price paid for reported earnings
Forward_PE	Expected earnings valuation	Reflects market expectations for future earnings

Beta_Volatility	Market sensitivity and risk	Distinguishes defensive from higher-beta companies
52_Week_Change_Pct	Recent price momentum	Captures investor sentiment and market performance

3. Methodology

- Download the Kaggle dataset through its API endpoint and read the CSV directly from the downloaded ZIP archive.
- Select the seven financial features and preserve the original company identifiers for later interpretation.
- Clip every feature at its 1st and 99th percentiles to reduce sensitivity to extreme values.
- Standardise each feature to a mean of zero and standard deviation of one using StandardScaler.
- Fit K-Means models for k = 2 to 15 with n_init = 15 and random_state = 41.
- Calculate the silhouette score for each candidate k and select the k with the highest score.
- Fit the final five-cluster model and calculate each cluster's mean standardised financial profile.
- Export company assignments, cluster profiles, silhouette scores and the heatmap to the current working directory.

Why standardisation is necessary

K-Means minimises squared Euclidean distance. Market capitalisation is measured in billions, while ratios such as beta and P/E use much smaller numerical scales. Without standardisation, market capitalisation would dominate the distance calculation. A standardised value above zero means the cluster is above the sample average for that feature; a value below zero means the cluster is below the sample average.

Model selection

The best solution is k = 5 with a silhouette score of 0.353. Silhouette scores range from −1 to +1: larger values indicate that observations are closer to their own cluster than to neighbouring clusters. A score of 0.353 suggests a meaningful structure, but also substantial overlap among technology-company profiles. This is reasonable because firms can share similar valuation or risk characteristics even when their industries and business models differ.

4. Cluster Profile Results

The heatmap below reports the mean standardised score for each financial feature by cluster. Red cells indicate above-average values, blue cells indicate below-average values, and values near zero indicate performance close to the sample average.

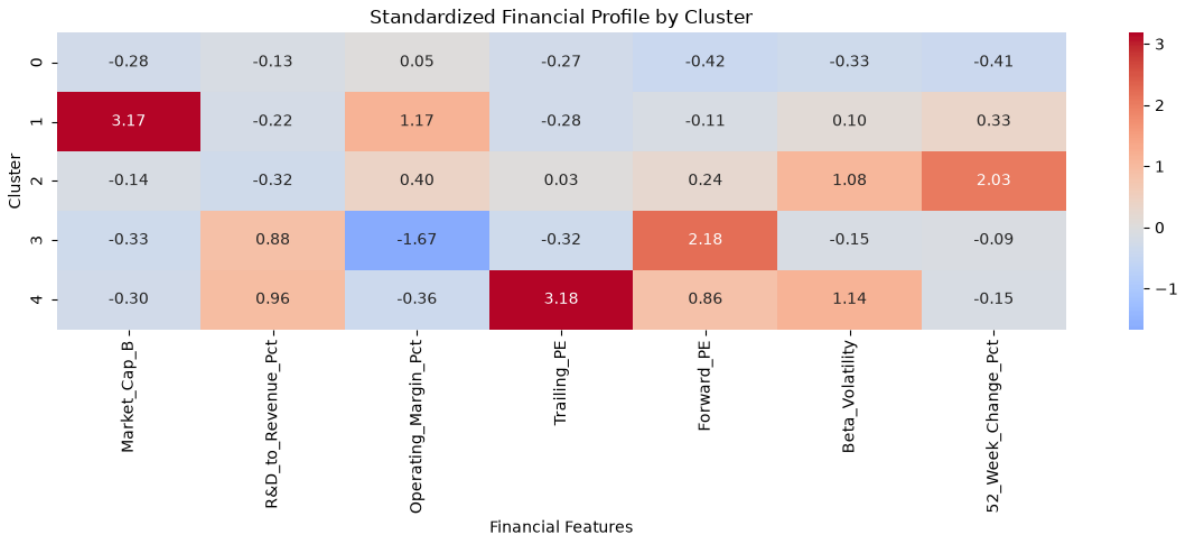


Figure 1. Standardised financial profile by cluster. Values are cluster-level z-score means.

Cluster	Suggested label	Most distinctive evidence	Investment interpretation
0	Lower-valuation, low-momentum companies	Market cap −0.28; trailing P/E −0.27; forward P/E −0.42; beta −0.33; momentum −0.41	Smaller, less expensive and more defensive companies, but with weak recent price performance. Operating margin is close to average.
1	Mega-cap profitable leaders	Market cap +3.17; operating margin +1.17; momentum +0.33	Exceptionally large and profitable firms. Valuation is below average and risk is close to average, suggesting established market leaders rather than highly speculative growth names.
2	High-momentum, higher-risk companies	Momentum +2.03; beta +1.08; operating margin +0.40	Companies with the strongest recent price performance and above-average market risk. Valuation is only modestly above average, while R&D intensity is below average.
3	R&D-intensive, low-profitability companies	R&D +0.88; forward P/E +2.18; operating margin −1.67	Innovation-heavy companies with very weak operating profitability and expensive forward valuation. Trailing P/E is below average, potentially reflecting unusual or unstable current earnings.
4	Expensive, high-beta innovation companies	Trailing P/E +3.18; R&D +0.96; forward P/E +0.86; beta +1.14	High valuation, high innovation spending and above-average risk, combined with below-average margins and slightly negative momentum. This group appears expectation-sensitive.

5. Detailed Interpretation

Cluster 0: Lower-valuation, low-momentum companies

Cluster 0 is below average in size, R&D intensity, both valuation multiples, beta and 52-week price performance. Operating margin is almost average at +0.05. This profile is consistent with smaller or mature technology firms that are not receiving strong investor enthusiasm. The low forward P/E and low beta may appeal to value-oriented or defensive investors, but the negative momentum suggests that company-specific catalysts and earnings quality require closer review.

Cluster 1: Mega-cap profitable leaders

Cluster 1 is dominated by a market-capitalisation score of +3.17, far above every other cluster, and an operating-margin score of +1.17. Momentum is moderately positive at +0.33, while trailing and forward P/E are below average. Economically, this is the strongest quality-and-scale cluster. These companies may benefit from durable competitive advantages, operating leverage and established cash-generating business models. The below-average P/E readings also imply that high size and profitability do not automatically correspond to the most expensive valuations in this sample.

Cluster 2: High-momentum, higher-risk companies

Cluster 2 has the strongest 52-week momentum score at +2.03 and an above-average beta of +1.08. Operating margin is also positive at +0.40, while valuation is only slightly above average. This cluster may contain market favourites whose prices have risen rapidly alongside solid profitability. The combination of strong momentum and high beta can support tactical opportunities, but it also increases reversal risk if earnings or market expectations weaken.

Cluster 3: R&D-intensive, low-profitability companies

Cluster 3 combines above-average R&D intensity (+0.88) with the lowest operating margin (−1.67) and the highest forward P/E (+2.18). This profile suggests companies investing heavily for future growth while current operating economics remain weak. The discrepancy between below-average trailing P/E and very high forward P/E should be investigated at company level because P/E ratios can become difficult to interpret when earnings are small, volatile or affected by one-off items.

Cluster 4: Expensive, high-beta innovation companies

Cluster 4 has the highest trailing P/E score (+3.18), together with high R&D (+0.96), above-average forward P/E (+0.86) and high beta (+1.14). Operating margin is below average and momentum is slightly negative. Investors are therefore paying a substantial valuation premium despite weaker margins and soft recent price performance. The cluster may contain long-duration growth companies whose valuations are particularly sensitive to interest rates, execution risk and revisions to future earnings expectations.

6. Limitations and Model Risks

Limitation	Why it matters / improvement
Moderate silhouette score	A score of 0.353 indicates overlap. Treat cluster labels as broad segments, not definitive classifications.
K-Means shape assumption	K-Means works best with roughly spherical, similarly sized clusters. Compare with hierarchical clustering, Gaussian mixtures or DBSCAN.
P/E ratio instability	P/E can be misleading for companies with low or negative earnings. Consider EV/EBITDA, price-to-sales and free-cash-flow yield.
Cross-sectional snapshot	The dataset reflects one period. Repeating the analysis over time would show whether cluster membership is stable.

Feature dependence	Trailing and forward P/E are related; market cap and profitability may also be correlated. Review correlations or use PCA.
No cluster-size analysis	The report cannot assess concentration without the number of companies per cluster. Add counts and percentages.
No company-level validation	Cluster interpretation should be checked against company names, industries and business models.
Outlier clipping	Winsorisation improves stability but can compress genuinely important extreme observations.

7. Conclusion

The K-Means model converts a multidimensional technology-company dataset into five economically meaningful profiles. The clearest segments are the **mega-cap profitable leaders in Cluster** 1, the **high-momentum higher-risk companies in Cluster 2**, the **R&D-intensive low-profitability companies in Cluster 3**, and the **expensive high-beta innovation companies in Cluster 4**. Cluster 0 provides a contrasting lower-valuation, lower-risk and weaker-momentum profile. The project demonstrates how unsupervised learning can support peer screening, portfolio construction and risk analysis. However, the moderate silhouette score means the clusters should be interpreted as analytical starting points rather than rigid investment categories.

Appendix: Files Produced by the Python Workflow

- company_cluster_result.csv: ticker, company name, industry and assigned cluster.
- cluster_profile.csv: mean standardised feature values for each cluster.
- silhouette_scores.csv: silhouette score for every tested number of clusters.
- cluster_profile_heatmap.png: visual comparison of cluster-level feature profiles.